User Guide

Introduction

The **A Real-Time Explainable Platform for Employee Attrition Prediction and Sentiment Analysis** is a web-based application developed to support employee attrition prediction and employee sentiment analysis using deep learning models. The application provides explainable predictions so that users can understand the factors influencing each result.

The application contains two main modules:

- **Employee Attrition Prediction**
- **Employee Sentiment Analysis**

Users can select different deep learning models for each task and compare their predictions. The application is hosted on Hugging Face Spaces and can be accessed through any modern web browser without installing additional software.

System Requirements

To use the application, users need:

- A computer, laptop, tablet, or smartphone.
- A stable internet connection.
- A modern web browser such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.

No software installation is required because the application runs entirely through the web browser.

Accessing the Application

Follow these steps to access the application.

1. Open a web browser.
2. Navigate to the application using the following URL:

https://huggingface.co/spaces/Ravishankarkota16/HR-Insights-Attrition-Sentiment

3. Wait for the application to load.
4. If the Hugging Face Space has entered sleep mode, it may take a few seconds to restart.
5. After loading, the home page displays the title **Real-Time Explainable Deep Learning Platform** together with two tabs:
    - Attrition Prediction
    - Sentiment Analysis

Using the Attrition Prediction Module

The Attrition Prediction module estimates whether an employee is likely to leave an organisation.

Step 1: Select the Module

Click the **Attrition Prediction** tab.

Step 2: Choose a Model

Select one of the available deep learning models:

- FT-Transformer
- Tabular ResNet

Step 3: Enter Employee Information

Complete all required input fields, including:

- City
- City Development Index
- Gender
- Relevant Experience
- Enrolled University
- Education Level
- Major Discipline
- Years of Experience
- Company Size
- Company Type
- Years Since Last New Job
- Training Hours

Step 4: Run the Prediction

Click the **Predict Attrition** button.

Step 5: Review the Results

The application displays:

- Predicted class (Stay or Leave)
- Prediction confidence
- Probability score

The feature importance section explains which employee characteristics contributed most to the prediction.

Using the Sentiment Analysis Module

The Sentiment Analysis module predicts whether an employee review expresses positive or negative sentiment.

Step 1: Select the Module

Click the **Sentiment Analysis** tab.

Step 2: Choose a Model

Select one of the available models:

- BiLSTM
- CNN-BiGRU-Attention

Step 3: Enter Review Text

Type or paste an employee review into the text box.

Example:

Great place to work with supportive managers.

Step 4: Analyse the Review

Click the **Analyse Sentiment** button.

Step 5: Review the Results

The application displays:

- Predicted sentiment
- Confidence score
- Sentiment probabilities

The Word Influence section highlights the words that most influenced the prediction.

- Green indicates positive influence.
- Red indicates negative influence.